Solved Model Paper Math FBISE Class 9s 9th Federal Board New Pattern 2027

50% MCQs & 50% Subjective

Sr no.	Question	A	B	C	D
1	The radical form of $2^{\frac{3}{4}} \times 27$ is:	X $3 \times \sqrt[3]{2^4}$	$3^3 \times \sqrt[4]{2^3}$	X $3^3 \times \sqrt[3]{2^4}$	$3^4 \times \sqrt[4]{2^4}$ X
2	The scientific notation of 150 million kilometres is:	1.5×10^6	1.5×10^8	1.5×10^{-6}	1.5×10^{-8}
3	The temperature in a freezer was $-\frac{9^\circ}{2}C$. It increased by $\frac{5^\circ}{2}C$. What is the new temperature?	$7^\circ C$	$2^\circ C$	$-2^\circ C$	$-7^\circ C$
4	For what values of x and y, $(x+y, 4x+1) = (3,9)$?	$x = 1$, $y = 2$	$x = 0.5$, $y = 8.5$	$x = 2$, $y = 1$	$x = 2$ $y = 7$
5	If $(x-1)$ is a factor of $x^3 - 5x^2 + ax + 6$, the value of a is:	-2	-1	0	2

(1) $\sqrt[4]{\ }$ $2^{3 \times \frac{1}{4}} = (2^3)^{\frac{1}{4}}$

(2) 1 Million = 10 Lac, = 1000000

$1 \sim 9$

$1.50,000,000.$

1.5×10^8 Km

C) $-\frac{9}{2} + \frac{5}{2} = \frac{-9+5}{2} = \frac{-4}{2} = -2^\circ$

		X	X		X
4	For what values of x and y, $(x+y, 4x+1) = (3,9)$?	$x = 1$, $y = 2$	$x = 0.5$, $y = 8.5$	$x = 2$, $y = 1$	$x = 2$ $y = 7$

5	If $(x-1)$ is a factor of x^3-5x^2+ax+6, the value of a is:	-2	-1	0	2

$x-1=0 \Rightarrow \boxed{x=1}$

$\boxed{x^3-5x^2-2x+6}$

$1-5+a+6=0$

$a+2=0$

$\Rightarrow a=-2$

$a+b$

6	If $x^2-11x+30=(x-a)(x-b)$, then value of $a+b$ is:	-30	-11	11	30

$=$

$\boxed{x^2-Sx+P=0}$

a, b

sum of roots

$a+b$

ab

$x^2-11x+30=x^2-bx-ax+ab$

$x^2-11x+30=x^2-(a+b)x+ab$

$a+b=11$

$ab=30$

7	The HCF of $6x^2y$ and $9xy^2$ is:	$3x^2y$	$3x^2y^2$	$3xy$	$18x^2y^2$
8	If $\sqrt{x^2+6x+9}=x+k^2$, the	-3	$+\sqrt{9}$	3	$+\sqrt{3}$

7	The HCF of $6x^2y$ and $9xy^2$is:	$3x^2y$	$3x^2y^2$	$3xy$	$18x^2y^2$
8	If $\sqrt{x^2+6x+9} = x + k^2$, the value of k is:	-3	$\pm\sqrt{9}$	3	$\pm\sqrt{3}$

7) $2 \times 3, \quad 3 \times 3$ $\quad 7, \quad y \times y$

$HCF = 3 \times x \times y = 3xy$

8) $\sqrt{(x^2+6x+9)} = x + k^2$

$\sqrt{(x)^2+2(x)(3)+(3)^2} =$

$\sqrt{(x+3)^2} = x + k^2$

$x + 3 = x + k^2$

$\sqrt{k^2} = \pm\sqrt{3}$

$k = \pm\sqrt{3}$

$x = -3$

$(x+3)$

$x^2 + 6x + 9 = (x+3)^2$

$x^2 + 6x + 9 = (x+3)$

9	The profit of a small business is modelled by $x^2 - 9x + 20$. For a breakeven point the factorized expression is:	$(x-4)(x-5)$	$(x-4)(x+5)$	$(x+4)(x-5)$	$(x+4)(x+5)$

$x^2 - 9x + 20 = 0$

$x = 5$

$x = 4$

$(x-5)(x-4)$

10	On solving $\frac{x}{2} - \frac{3}{2} \geq 2$, the value of x is:	$x \leq 5$	$x \geq 5$	$x \geq 7$	$x \leq 7$

$\frac{x-3}{2} \geq 2$

$x - 3 \geq 4 \Rightarrow x \geq 7$

11	Which value of y satisfies the equation $0.6y + 1.2 = 3$?	0.3	1.08	3	7

?

$0.6(y) + 1.2 = 3$

$0.6y = 3 - 1.2 = 1.8$

$0.6y = 3 - 1.2$

$0.6y = 1.8$

$y = \frac{1.8}{0.6} = 3$

12	What is the midpoint of the line segment joining A (2, 4) and B (6, 10)?	(2,3)	(4,7)	(6,5)	(8,14)

x, y

$A(2, 4)$ $B(6, 10)$

$x = \frac{2+6}{2},$ $y = \frac{4+10}{2}$

$A(x_1, y_1)$ $B(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

slope

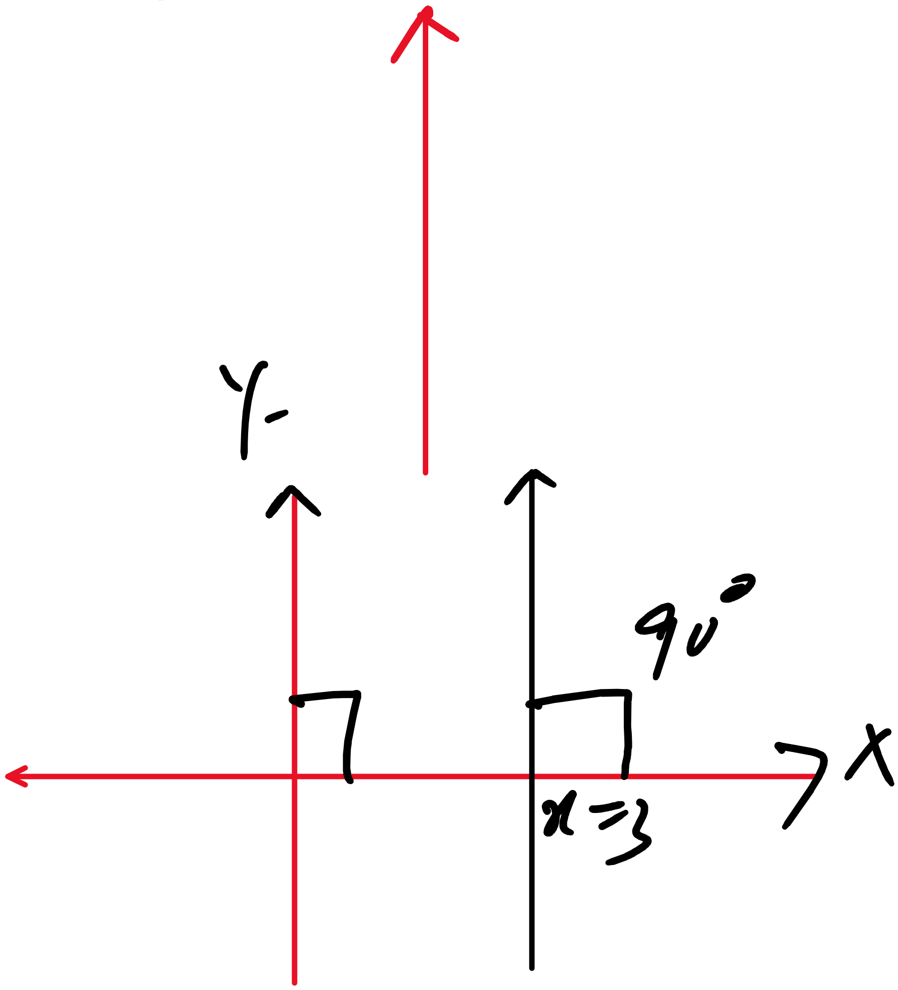
13	What is the gradient of a line passing through points (3, 7)	0	−2	$\frac{1}{-}$	undefined

13	What is the gradient of a line passing through points (3, 7) and (3, –5)?	0	−2	$\frac{1}{3}$	undefined

$\overset{x_1,\ y_1}{A(3,7)}$ $\overset{x_2,\ y_2}{B(3,-5)}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-5-7}{0} =$ undf

$\frac{-12}{0} \neq 0$ $\frac{0}{-12} = 0$ ✓



14	If $y = 2x + c$ passes through the point (3, 10), what is the numerical value of c?	−17	4	16	23

14	... the point (3, 10), what is the numerical value of c?		4	10	25

$\boxed{y = mx + c}$ → standard / straight line eqn

gradient/slope

y-intercept

$ax + by + c = 0$ → general form

$y = 2x + c$

x, y

(3, 10)

$10 = 2(3) + c$

$10 = 6 + c$

$c = 10 - 6 = 4$

$m_1 = 5$

$5x - \frac{1}{5} = -1$

15	The line $y = 5x + 1$ is perpendicular to another line. What is the gradient of the other line?	5	-5	$\frac{1}{5}$	$-\frac{1}{5}$

$m \rightarrow$ gradient/slope

$= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

$m_2 = -\frac{1}{m_1} = -\frac{1}{5}$ if

$\xrightarrow{m_1} l_1$

$\xrightarrow{m_2} l_2$

$l_1 \parallel l_2 \Rightarrow m_1 = m_2$

↳ parallel lines

↳ parallel lines

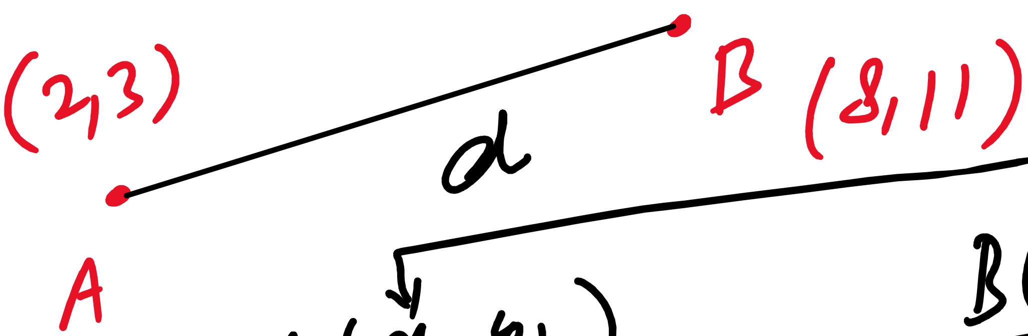
If $l_1 \perp l_2 \Rightarrow$ $m_1 m_2 = -1$

↳ perpendicular symbol.

$$m_2 = -\frac{1}{m_1}$$

$\sqrt{36+64} = 10 = \sqrt{6^2+8^2}$ $\quad \sqrt{(8-2)^2+(11-3)^2}$

16	Two schools are located at points (2, 3) and (8, 11) on a map. What is the distance between them?	100 units	14 units	10 units	5 units



$A(x_1, y_1)$ $\quad B(x_2, y_2)$

$\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$d = \sqrt{(x_2 - x_1)^2 + (y ...}$ distance b/w two points

$$d = \sqrt{(2-8)^2 + (3-11)^2}$$

$$d = \sqrt{(-6)^2 + (-8)^2} = \sqrt{36+64} = \sqrt{100}$$

$= 10$

17	On a toy model car, the length of the door is 3 cm. If the actual car is exactly 40 times larger than the model, what is the length of the door on the actual car?	43 *cm*	80 *cm*	120 *cm*	1200 *cm*

door size in model = 3cm

Actual length = 40 × 3cm

= 120cm

18	A bakery makes two sizes of similar cube-shaped cakes. The larger cake has sides that are twice as long as the smaller cake. If the small cake requires 1 cup of butter, how many cups of butter are needed for the larger cake?	2 cups	4 cups	8 cups	16 cups

Large smale

Volume

2 : 1

Cube-shaped cakes

Larger cake = 2 × smaller cake

$$\text{Volum} = \frac{l \times w \times h = cm^3}{} = l^3$$

Butter

$$\text{cups} = 1 \times 2^3 = 8 \text{ cups}$$

8 × 1 butter

2 : 1

# of butter cups requ for

larger size $= 2^3 \times 1$

larger size $= 2^{x}$ ...

$= 8$

19	Two intersecting roads form supplementary angles where one angle is three times the other. What is the measure of the smaller angle?	22.5°	45°	90°	135°

Sum of Angles $= 180° \Rightarrow$ supplementary

$3x$, x

2nd Angle   1st Angle

$$3x + x = 180°$$

$$4x = 180$$

$$x = \frac{180}{4} = 45°$$

20	Two sides of a triangle have slopes 1 and -1. What is the angle between them?	45°	90°	135°	180°

$1 \times -\frac{1}{1} = -1$

$m_1 \times m_2 = -1$

21	Two similar triangles have corresponding sides in the ratio 1: 3. What is the ratio of their areas?	1 : 6	6 : 1	9 : 1	1 : 9

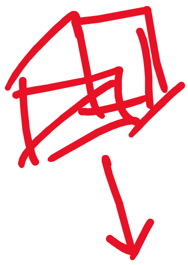
1 : 3

$\text{Area} = (1:3)^2 = 1:9$

22	A cuboid has volume 1000 cm^3. It is enlarged by a scale factor of 2. What is the volume of the enlarged cuboid?	500 cm^3	2000 cm^3	4000 cm^3	8000 cm^3

Volume = 1000 cm^3

Scaling factor = 2

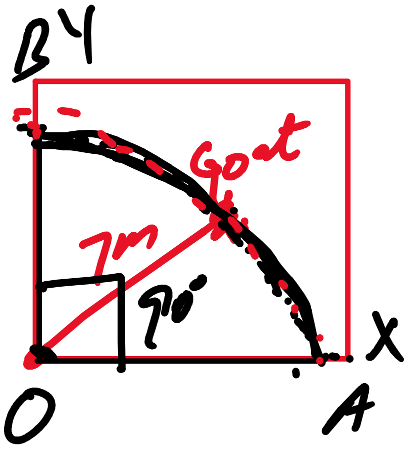


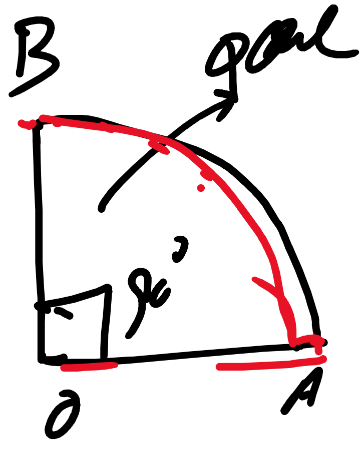
Volume of large cube $= 2^3 \times 1000$

$= 8000 \, cm^3$

23	A goat is tied to a corner of a square field with a 7m rope. The locus of points it can graze is a:	Quarter circle	Semi-circle	Full circle	Square

$$\frac{360^\circ}{90^\circ} = 4$$





24	A dog is tied to a fixed pole in a park with a 5 m rope. The set of all points the dog can reach is a:	Circle	Semi-circle	Square	Rectangle

25	135° in circular system (radians) is:	$\frac{3\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{4}$

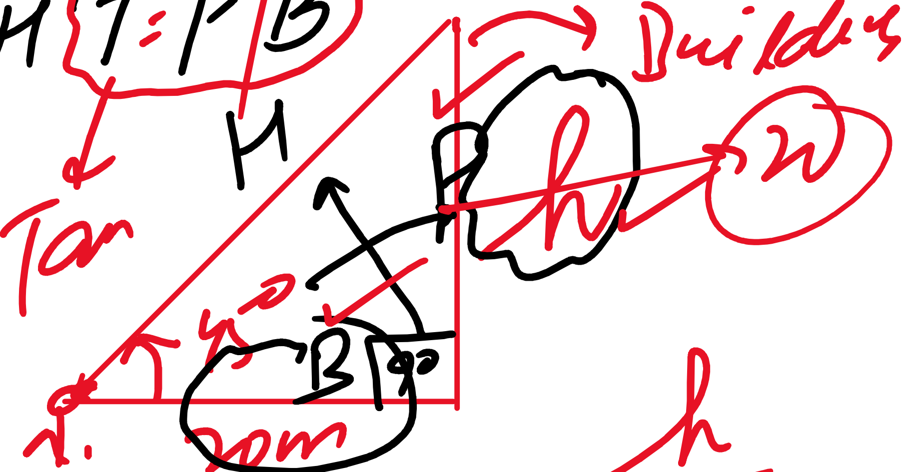
$$135 \times \frac{\pi}{180} = \frac{3\pi}{4}\text{rad}$$

$$\frac{3\pi}{4} \times \frac{180}{\pi} = 135^\circ$$

26	A person stands 20 m from a building. The angle of elevation to the top is 45°. What is the height of the building?	11.5 m	14.1 m	20 m	34.6 m

$S = P/H \quad C = B/H \quad T = P/B$

sin cos Tan



$$\tan 45^\circ = \frac{P}{B} = \frac{h}{20}$$

$$\frac{h}{20} = 1 \Rightarrow h = 20$$

27	The value of $\cos^2 100° + \sin^2 100°$ is:	0	0.81	1	1.15

$$\sin^2\theta + \cos^2\theta = 1$$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \csc^2\theta$

$\sec\theta = \frac{1}{\cos\theta}$, $\csc\theta = \frac{1}{\sin\theta}$

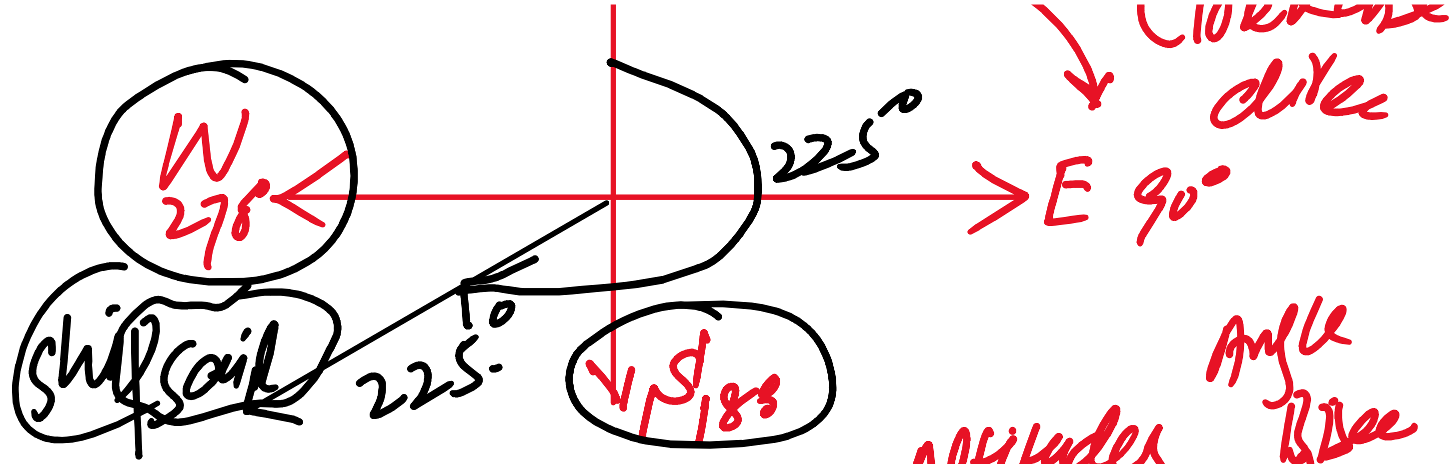
$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta} = \cot\theta = \frac{1}{\tan\theta}$

28	A ship sails from port on a bearing of 225°. In which direction is it moving?	North-East	South-East	South-West	North-West

$360°$ $0°$ N

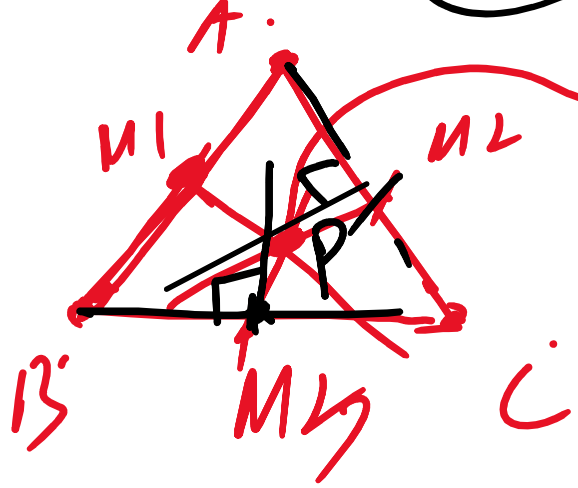
Clockwise



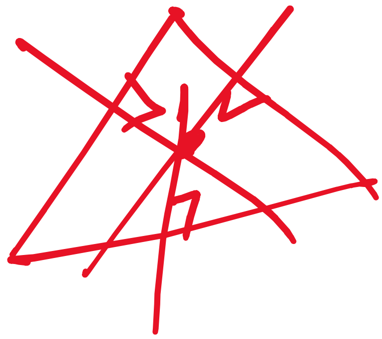
29	In a triangle, the three medians meet at a point called the:	Circumcenter	Orthocenter	Centroid	Incentre

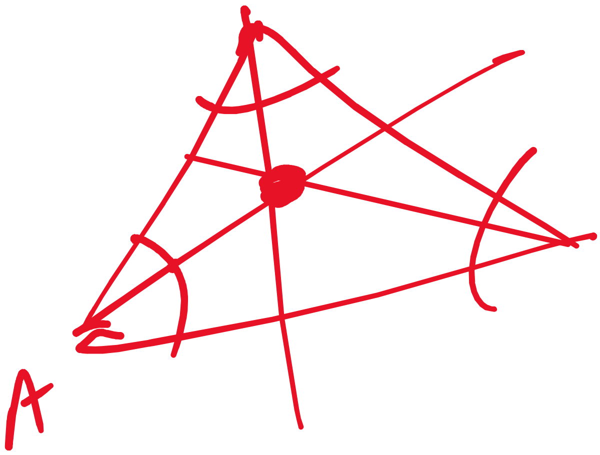
Altitudes

Angle Bisec



Perpendicular Bisector



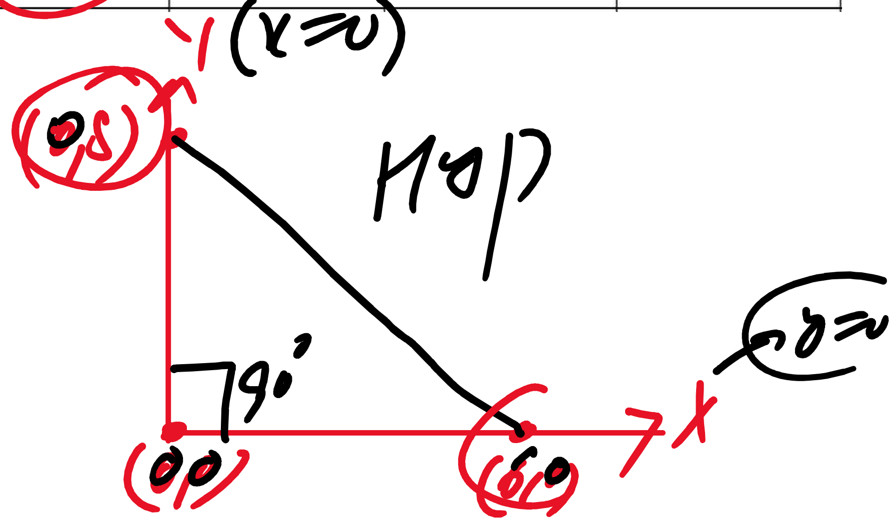


30	The vertices of a right-angled triangle are (0, 0), (6, 0), and (0, 8). What is the midpoint of the hypotenuse?	(3, 4)	(6, 8)	(3, 0)	(0, 4)

30	(0, 8). What is the midpoint of the hypotenuse?	(3, 4)	(6, 8)	(3, 0)	(0, 4)

$\left(\frac{0+6}{2}, \frac{8+0}{2}\right)$

$(3, 4)$



31	What is the probability of getting M in MATHEMATICS?	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{11}$	$\frac{2}{11}$

11

$\frac{2}{11}$

$0 \le P(E) \le 1$

32	A fair coin is tossed 100 times. The expected number of Tails to occur is:	0	1	50	100

Fair Coin

$P(H) = \frac{1}{2}$

Head Tail

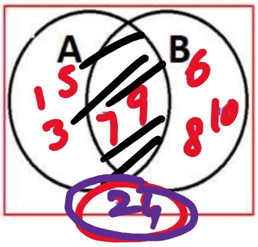
$f = 100$

$P(T) = \frac{1}{2}$

Expected $f(T)$

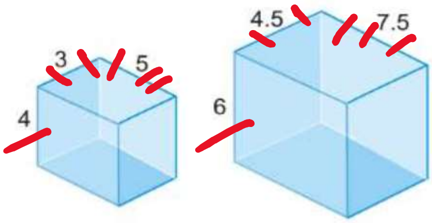
$100 \times \frac{1}{2} = 50$

Expected E(T) = $100 \times \frac{1}{2} = 50$

	Scenario for MCQs 33 to 36	**The Venn diagram shows the elements of set A and set** A∩B = {7,9} A-B 							
33	Which of the elements belong only to set A?	{1,2,4,6, 7}	{1,3,5,7,9}	{1,3,5}	{7,9}	○	○	○	○
34	Which of the elements are in the intersection of sets A and B?	{1,3,5, ... ,10}	{7,9}	{1,3,5}	{6,8,10}	○	○	○	○
35	Which of the elements are not included in either set A or set B?	{1,2,3, ... ,10}	{1,3,5}	{2,4}	{6,8,10}	○	○	○	○
36	The elements of $A - B$ are:	{1,3,5}	{6,8,10}	{7,9}	{2,4}	○	○	○	○

A difference B

✓ A, NOT B

	Scenario for MCQs 37 to 40	**The given prisms are similar** 							
37	What is the linear scale factor from the smaller cuboid to the larger cuboid?	1.25	1.5	2.0	2.25	○	○	○	○
38	The total surface area of the smaller cuboid is 94 cm^2, what is the total surface area of the larger cuboid?	141 cm^2	211.5 cm^2	317.25 cm^2	423 cm^2	○	○	○	○
39	The volume of the smaller cuboid is 60 cm^3, what is the volume of the larger cuboid?	90 cm³	135 cm³	202.5 cm³	270 cm³	○	○	○	○
40	What is the ratio of the volume of the smaller prism to the volume of the larger prism?	1 : 1.5	1 : 2.25	1 : 3.375	1 : 4.5	○	○	○	○

1.5 : 1

volume of the larger prism.

1.5 : 1

Area = $94 \times (1.5)^2$

Larger Volume = $60 \times (1.5)^3$ cm³

= 202.5 cm³

Smaller Volume = 60

$\frac{60 \text{ cm}^3}{202.5}$

$\frac{60}{60} : \frac{202.5}{60}$

1 : 3.375

C-I Class Intervals f → freq

	Scenario for MCQs (41-44)	The table below shows weight distribution (in kg) of students in a class.							
41	What is the size of class interval?	10	11	12	15	○	○	○	○
42	What is the value of $\frac{\Sigma f}{2}$?	20	22	25	26	○	○	○	○
	What is the cumulative								

C-I	30-40	40-50	50-60	60-70	70-80
f	6	10	16	12	6

	interval?								
42	What is the value of $\frac{\Sigma f}{2}$?	20	22	25	26	O	O	O	O
43	What is the cumulative frequency for the class 50 – 60?	16	26	32	34	O	O	O	O
44	Which is the median class?	40 – 50	50 – 60	60 – 70	30 – 40	O	O	O	O

$\frac{\Sigma f}{2} = 25$

$41 \Rightarrow \quad 40 - 30 = 10$

$$\frac{\Sigma f}{2} = \frac{6+10+16+12+6}{2} = \frac{50}{2} = 25$$

$$CF = 6 + 10 + 16 = 32$$

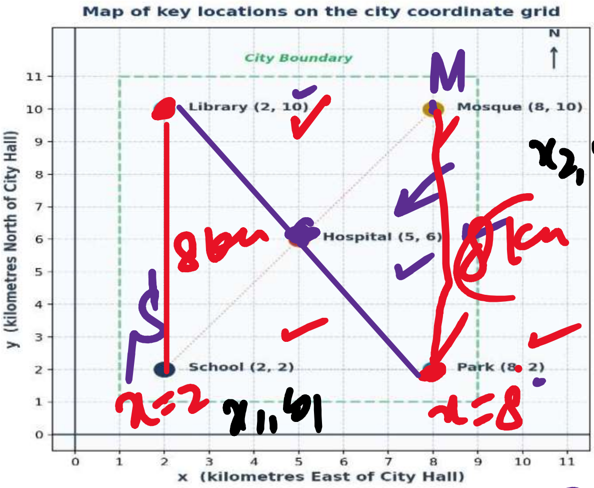
$$\Sigma f = 50 = N$$

Mid value $= \frac{\Sigma f}{2} = \frac{N}{2} = 25$

Study the given map and answer the following questions.

Scenario for MCQs (45-48)

$d = \sqrt{(8-2)^2 + 0^2} = 6\text{ km}$



$8 - 2 = 6$

$\frac{8+2}{2} = \frac{10}{2} = 5$

$\frac{2+10}{2} = \frac{12}{2} = 6$

45	What is the distance between School and Park?	4 km	6 km	8 km	10 km	○	○	○	○
46	What is the midpoint of the line segment joining Library and Park?	(4, 5)	(5, 5)	(5, 6)	(6, 5)	○	○	○	○
47	What is the slope of the line joining two points (school and mosque)?	$\frac{4}{3}$	$\frac{3}{4}$	$\frac{6}{5}$	$\frac{5}{6}$	○	○	○	○
48	Which pair of locations on the map is exactly 8 km apart?	School & Hospital	Park & Hospital	Park & Mosque	Library & Hospital	○	○	○	○

$m = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$

$\frac{10-2}{8-2}$

$m = \frac{8}{6} = \frac{4}{3}$

$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

(x_2, y_2)

Δy

(x_1, y_1) Δx

$90°$

m = undefined

m = 0 $0°$

l_1

m → -ve

$y_2 - y_1$

l_1

m +ve

left → RS